

Shuqi Zhong
116777748
Shuqzhong
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ISE331-HW4

9.1 A developer in your company is building a new application and has asked you if she should use TCP- or UDP-based communications. Provide her with a brief discussion of the advantages and disadvantages of each protocol.

TCP and UDP are two transport protocols used for communication, but they have different purposes. TCP is connection-oriented, which means it sets up a connection before sending data and guarantees that the data arrives correctly and in order. This makes it reliable, but also slower because of extra overhead like acknowledgments and retransmissions. On the other hand, UDP is connectionless and much faster because it does not guarantee delivery or order of packets. However, this also means some data can be lost. In general, TCP is better for applications where reliability matters, like file transfer, while UDP is better for real-time applications like video streaming or gaming.

9.3 Describe three basic types of network topologies and provide a sample diagram of each type.

Three basic types of network topologies are bus, star, and ring. In a bus topology, all devices share a single communication line, which is simple and low cost, but if the main cable fails, the whole network stops working. In a star topology, all devices connect to a central hub or switch, which makes it easy to manage, but if the central device fails, the network goes down. In a ring topology, devices are connected in a circle and data travels in one direction, but if one device fails, it can affect the entire network. Each topology has its own advantages and disadvantages depending on the use case.

9.4 Describe the TCP three-way handshake process.

The TCP three-way handshake is used to establish a connection between a client and a server. First, the client sends a SYN packet to the server to request a connection. Then, the server responds with a SYN-ACK packet to acknowledge the request and indicate it is ready. Finally,

the client sends an ACK packet back to the server to confirm the connection. After these three steps, the connection is established and data can be transmitted reliably between the two sides.

9.5 Your boss wants to know how subnetting works. Provide her with a brief description and be sure to include an example to illustrate how subnetting works.

Subnetting is the process of dividing a larger network into smaller networks to improve organization and efficiency. It works by using a subnet mask to separate the network portion and the host portion of an IP address. For example, a network like 192.168.1.0/24 can be divided into two smaller subnets, such as 192.168.1.0/25 and 192.168.1.128/25. Each subnet has fewer hosts, which helps reduce network congestion and improve security. Subnetting is commonly used in large networks to better manage IP addresses and control traffic.

CMK.17.1 Explain how a TLS connection is established.

A TLS connection is established through a process called a handshake. First, the client sends a “Client Hello” message that includes supported encryption methods and a random value. Then, the server replies with a “Server Hello” message, selects a cipher suite, and sends its certificate. The client verifies the server’s certificate to make sure it is trustworthy. After that, both sides generate a shared secret key using key exchange methods. Once the handshake is complete, they use this shared key to encrypt all future communication. This process ensures that the connection is secure and private.

CMK.17.2 What is the purpose of random nonces in the SSL/TLS handshake?

Random nonces in the SSL/TLS handshake are used to improve security by adding randomness to the key generation process. These values are generated by both the client and the server and are used to create unique session keys. This helps prevent replay attacks because each session will have different values and cannot be reused by an attacker. Nonces also make the encryption stronger by ensuring that the keys are unpredictable. Overall, they play an important role in keeping the communication secure.